

Additional file 1 — Late S-wave energy in Utah FORGE downhole microearthquakes is not reproduced by smooth elastic models: 3-D tests against stochastic small-scale heterogeneity

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This file contains the diagnostics referred to in the main text. Figures S3, S4 and S10–S12 and Table S14 come from the original analysis (fixed $\sigma = 0.13$ for the exponential and Gaussian media, 150 events); they are retained for completeness and are superseded in the main text by the per-event test on all events, the common-grid comparison and the cluster-bootstrap coherence analysis.

Supplementary figures

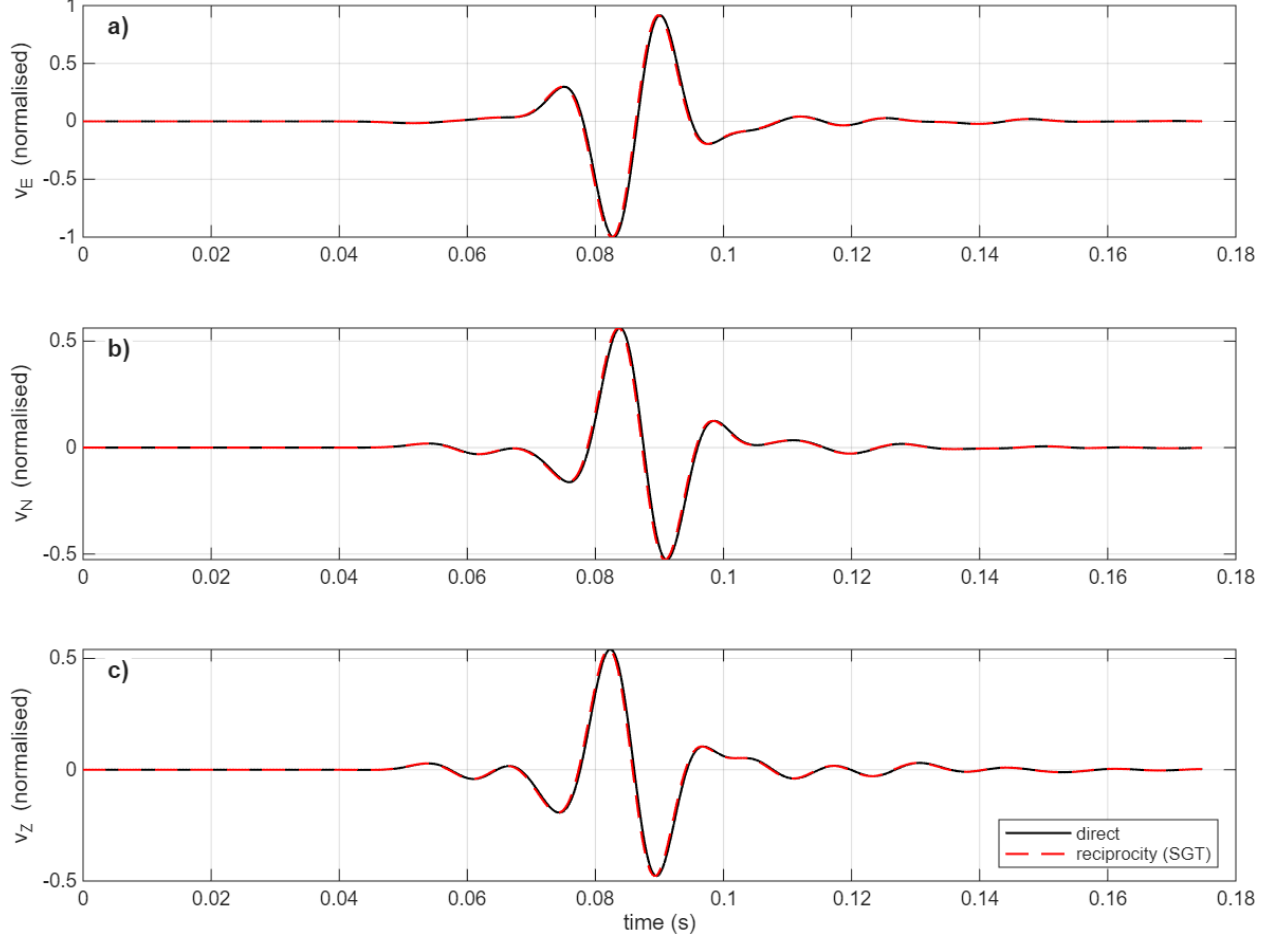


Fig. S1 Reciprocity validation in a heterogeneous VTI pink-noise medium ($\sigma = 0.045$, $\varepsilon = 0.05$, $\delta = 0.02$, $\gamma = 0.05$). a–c) East, north and vertical velocity at a sensor from a direct double-couple simulation (black) and from the strain-Green's-tensor reciprocity synthetic (red dashed). Correlation and amplitude ratio are 0.995.

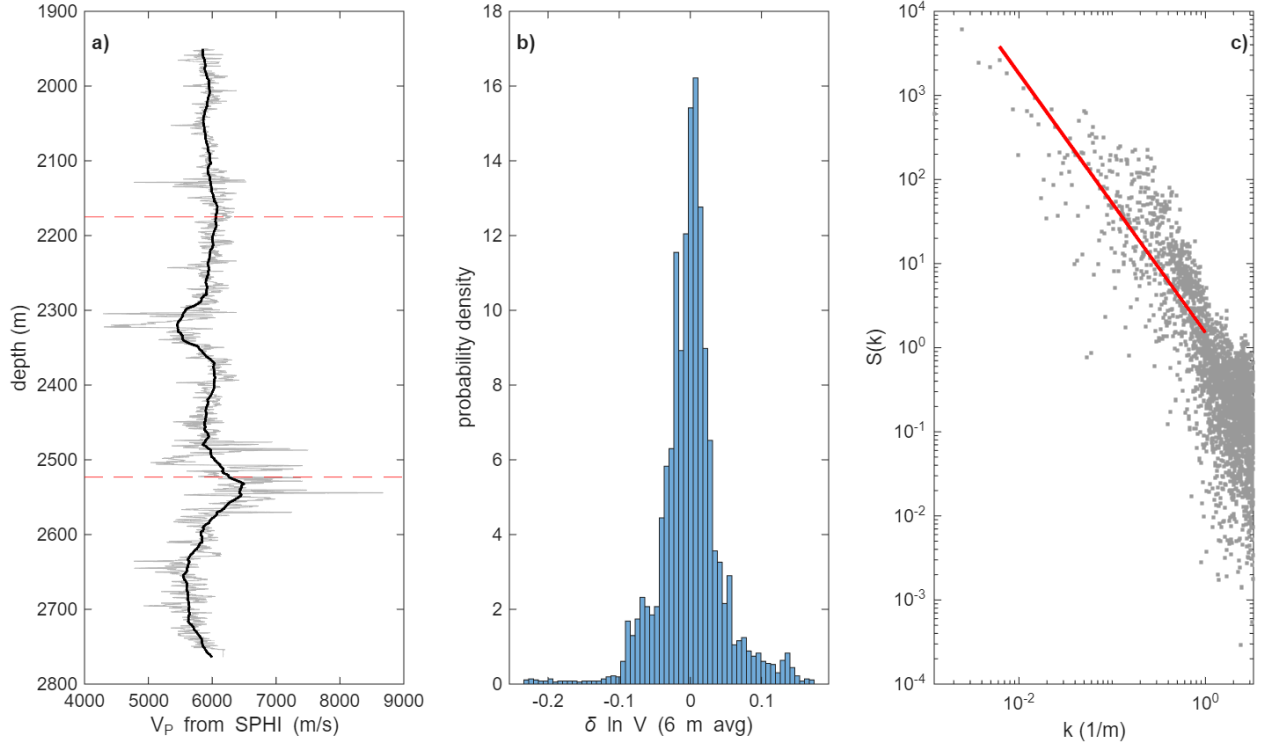


Fig. S2 The 56-32 sonic log. a) Sonic-derived V_p (grey) and its 50-m block average (black); dashed lines mark the depths of sensors A and B. b) Distribution of the detrended, 6-m-averaged $\delta \ln V$. c) Periodogram of $\delta \ln V$ with the original power-law fit over 1–200 m. Slope uncertainty is given in Table S15 and main-text Fig. 9.

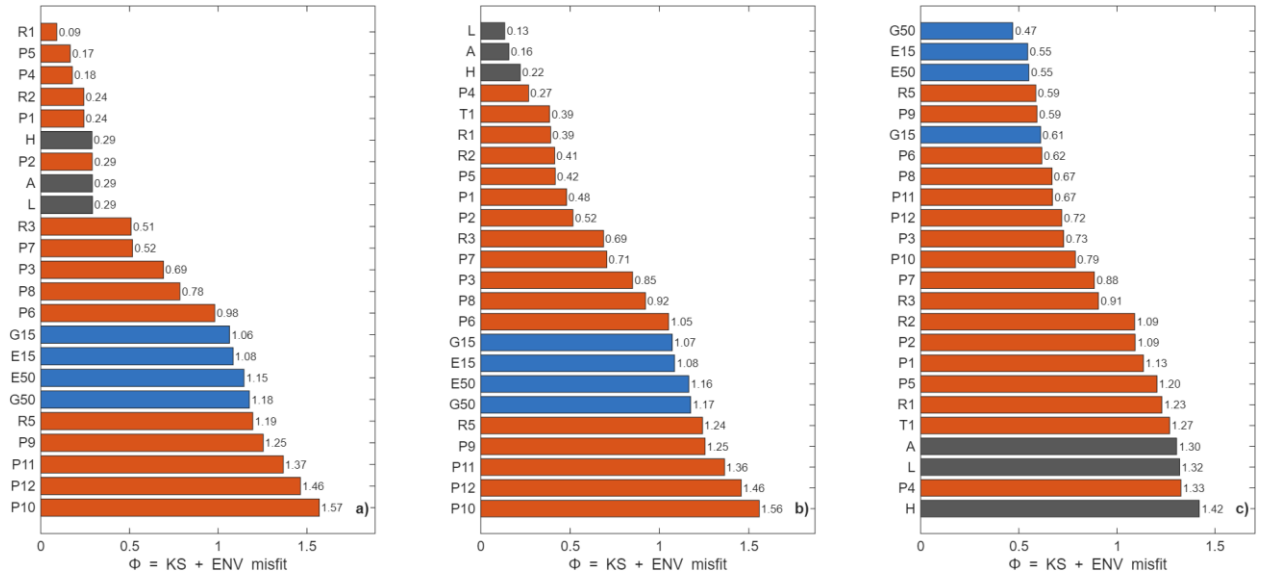


Fig. S3 Original statistical-misfit ranking (Φ) for a) the synthetic pink truth T1 ($\sigma = 0.045$), b) the layered-VTI truth T2 and c) FORGE (original model set: exponential and Gaussian media only at $\sigma = 0.13$). Grey: smooth media; orange: pink; blue: exponential and Gaussian. Lower is better. Superseded for FORGE by main-text Fig. 8.

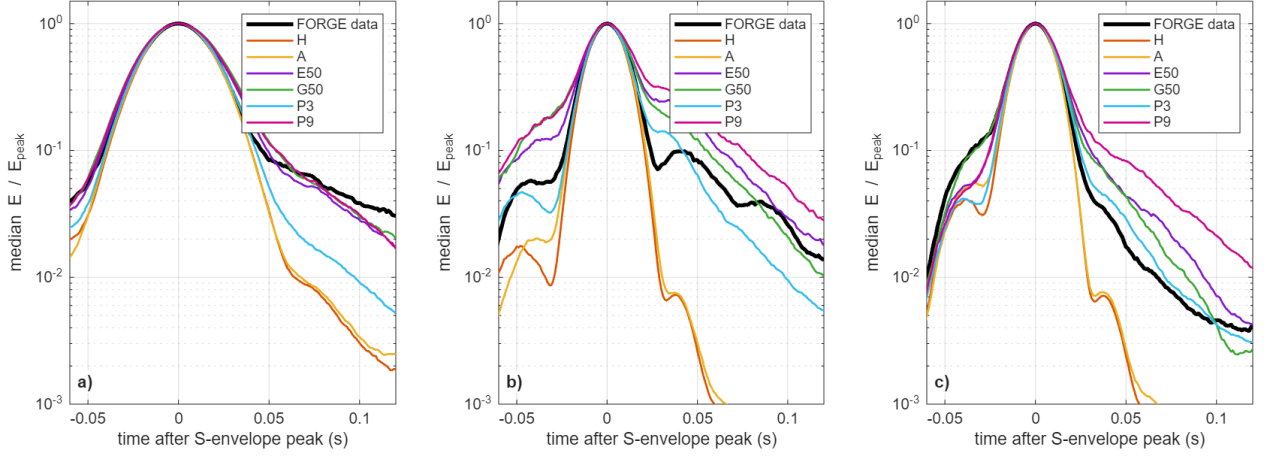


Fig. S4 Median S-coda envelopes aligned on the S peak and normalised by it: a) sensor A, 20–40 Hz; b) sensor A, 40–80 Hz; c) sensor B, 40–80 Hz. Black: FORGE; H homogeneous; A VTI; E50/G50 exponential/Gaussian ($a = 50$ m, $\sigma = 0.13$); P3/P9 pink ($\sigma = 0.09/0.18$).

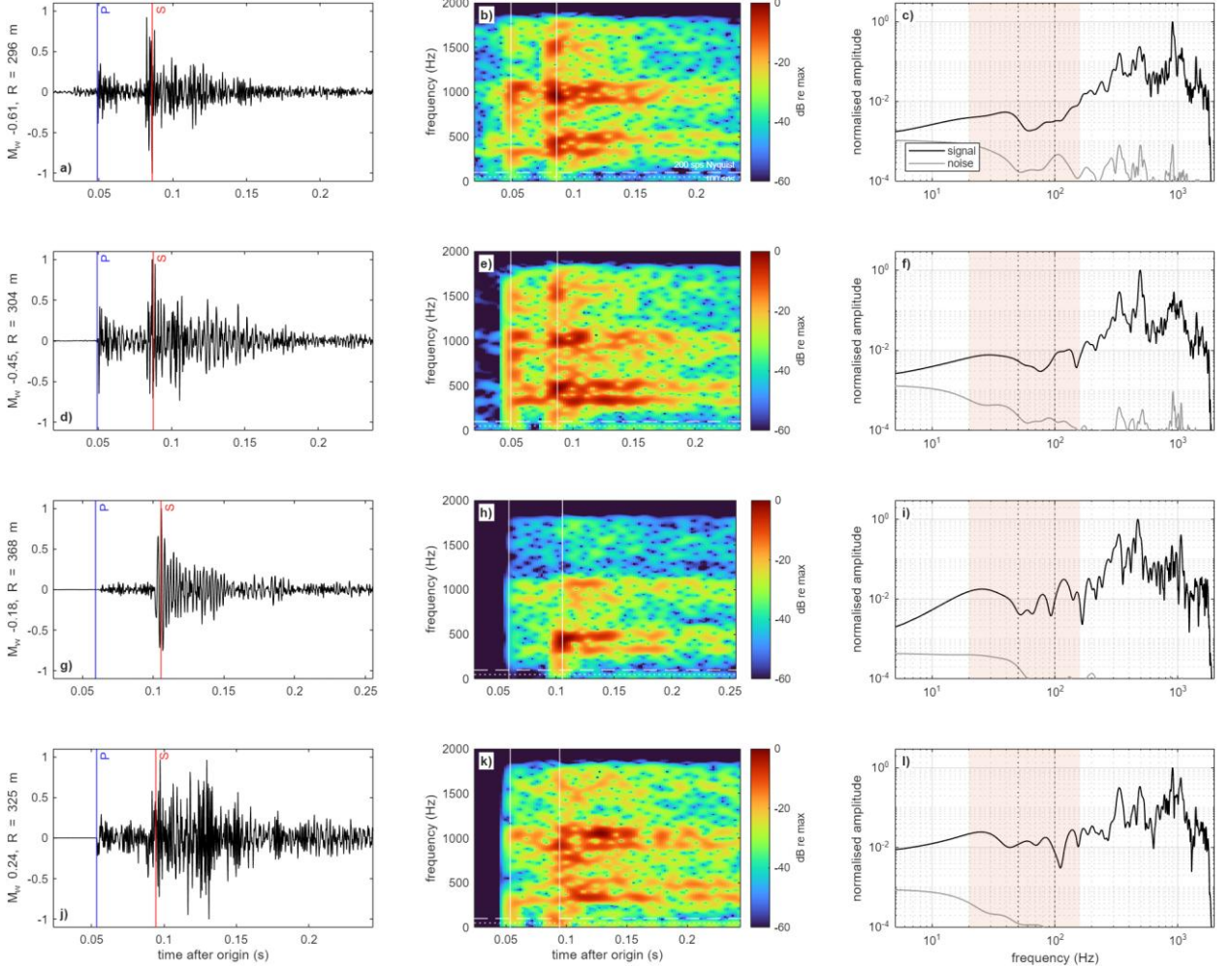


Fig. S5 Four example events (magnitude -0.61 to $+0.24$) at sensor B, 4 kHz. a, d, g, j) Horizontal velocity (> 5 Hz) with P (blue) and S (red). b, e, h, k) Spectrograms to 2 kHz on a common dB scale relative to each event's maximum; white dashed and dotted lines mark 100 and 50 Hz, the Nyquist frequencies of 200 and 100 sample/s recording. c, f, i, l) S-window spectrum (black) and pre-event noise (grey); shaded: FD analysis band.

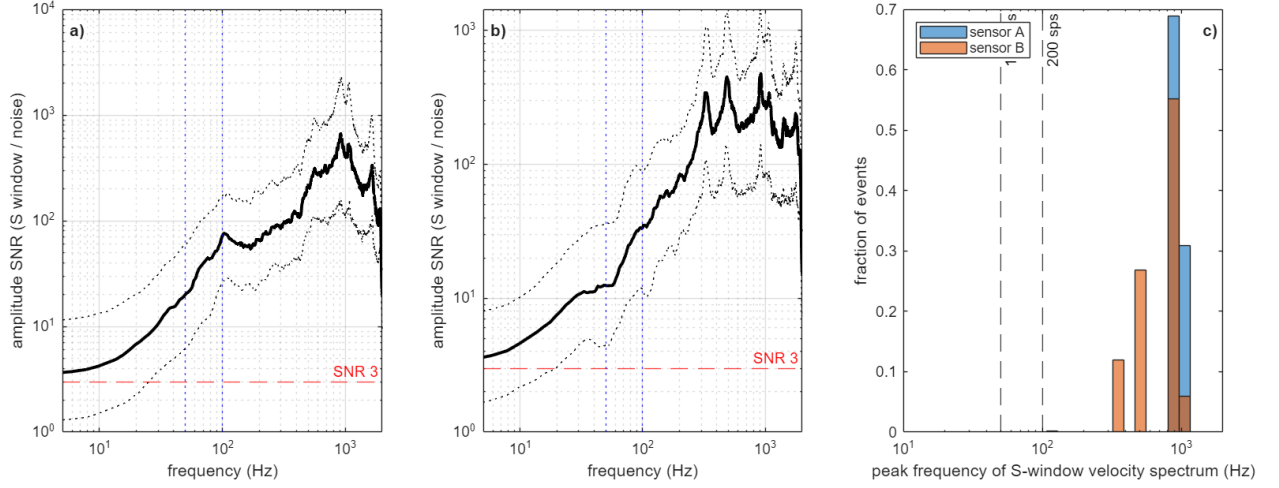


Fig. S6 a, b) Median (solid) and interquartile range (dotted) of the S-window amplitude SNR spectrum for sensors A and B; horizontal line at SNR = 3. c) Peak frequency of the S-window velocity spectrum for all events. The clustering near 330, 480 and 920 Hz indicates tool or coupling resonances; the share of recorded velocity energy above 100 Hz is therefore not a source or path property.

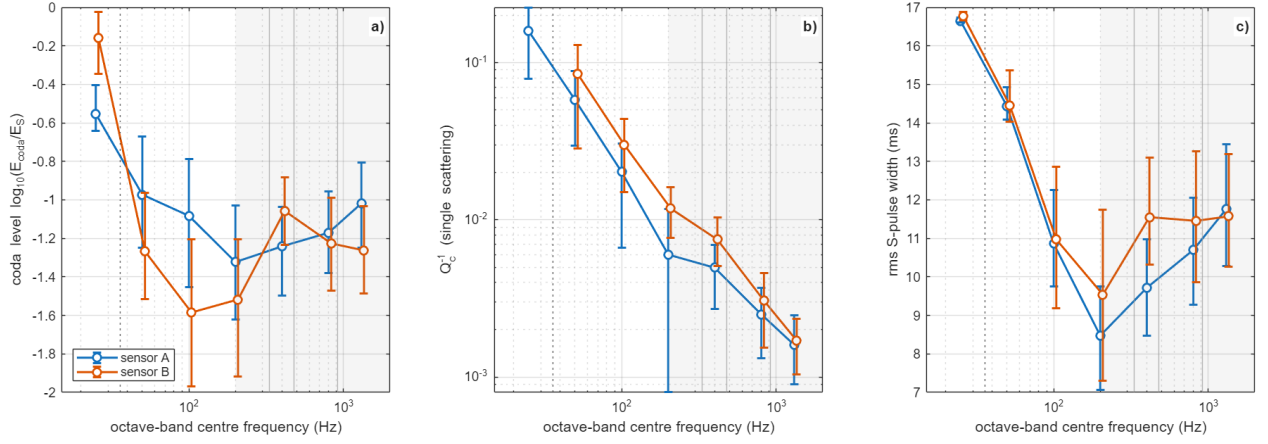


Fig. S7 FORGE coda measures in seven octave bands (25 Hz to 1.3 kHz): a) coda level, b) single-scattering Q_c^{-1} (finite-window diagnostic; at 25 Hz the window spans one to three periods), c) rms S-pulse width. Median and interquartile range for sensors A and B. Shaded (> 200 Hz): affected by the resonances.

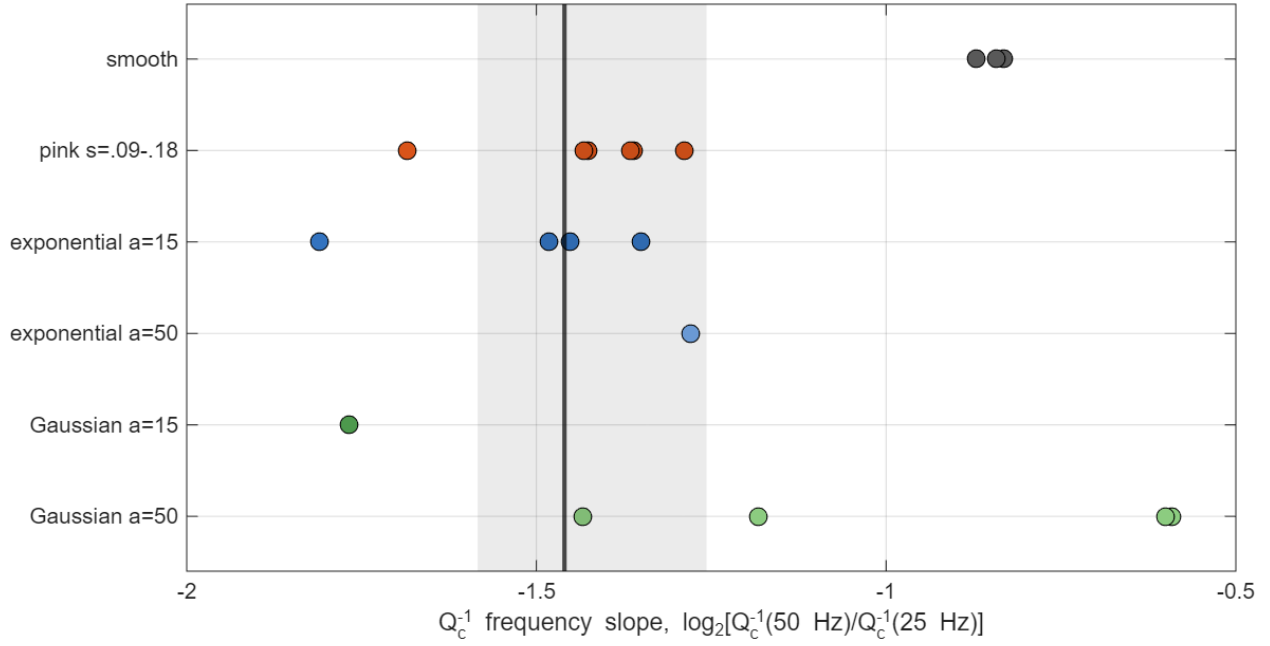


Fig. S8 Q_c^{-1} frequency slope $\log_2[Q_c^{-1}(50 \text{ Hz})/Q_c^{-1}(25 \text{ Hz})]$ at sensor A, FORGE (black line, grey 95 % bootstrap band) and simulated media (dots). A finite-window diagnostic: data and synthetics are measured identically, but the absolute values are not medium properties.

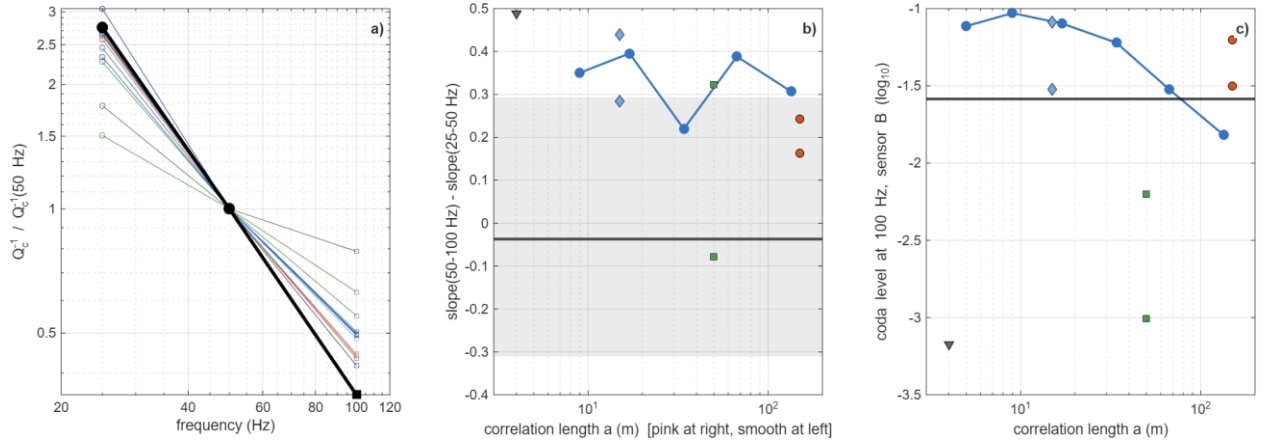


Fig. S9 Exploratory high-frequency series (2.5-m grid, sensor B, reduced volume, noise-free synthetics, $\sigma = 0.13$). a) Q_c^{-1} normalised at 50 Hz from the 25–50 Hz (sensor A, 5-m grid) and 50–100 Hz (sensor B, 2.5-m grid) slopes. b) Curvature of Q_c^{-1} versus correlation length. c) Coda level at 100 Hz versus correlation length; black line: FORGE. Grid, domain, sensor and event subset change together between the two octaves, so the comparison is not a pure frequency effect.

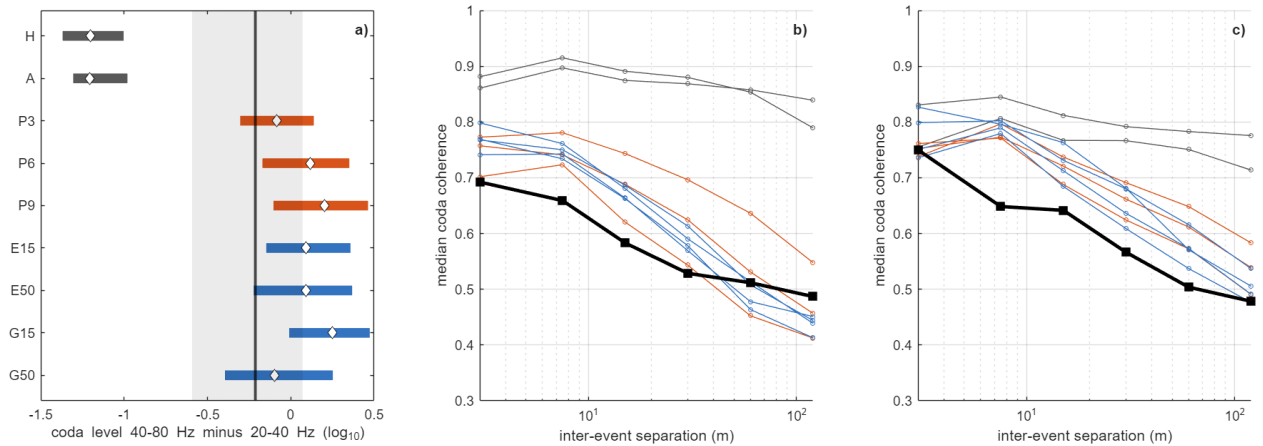


Fig. S10 Original D1 (coda level at 40–80 Hz minus 20–40 Hz, sensor A) and D2 (median coda coherence versus separation) without uncertainty; superseded by main-text Fig. 7.

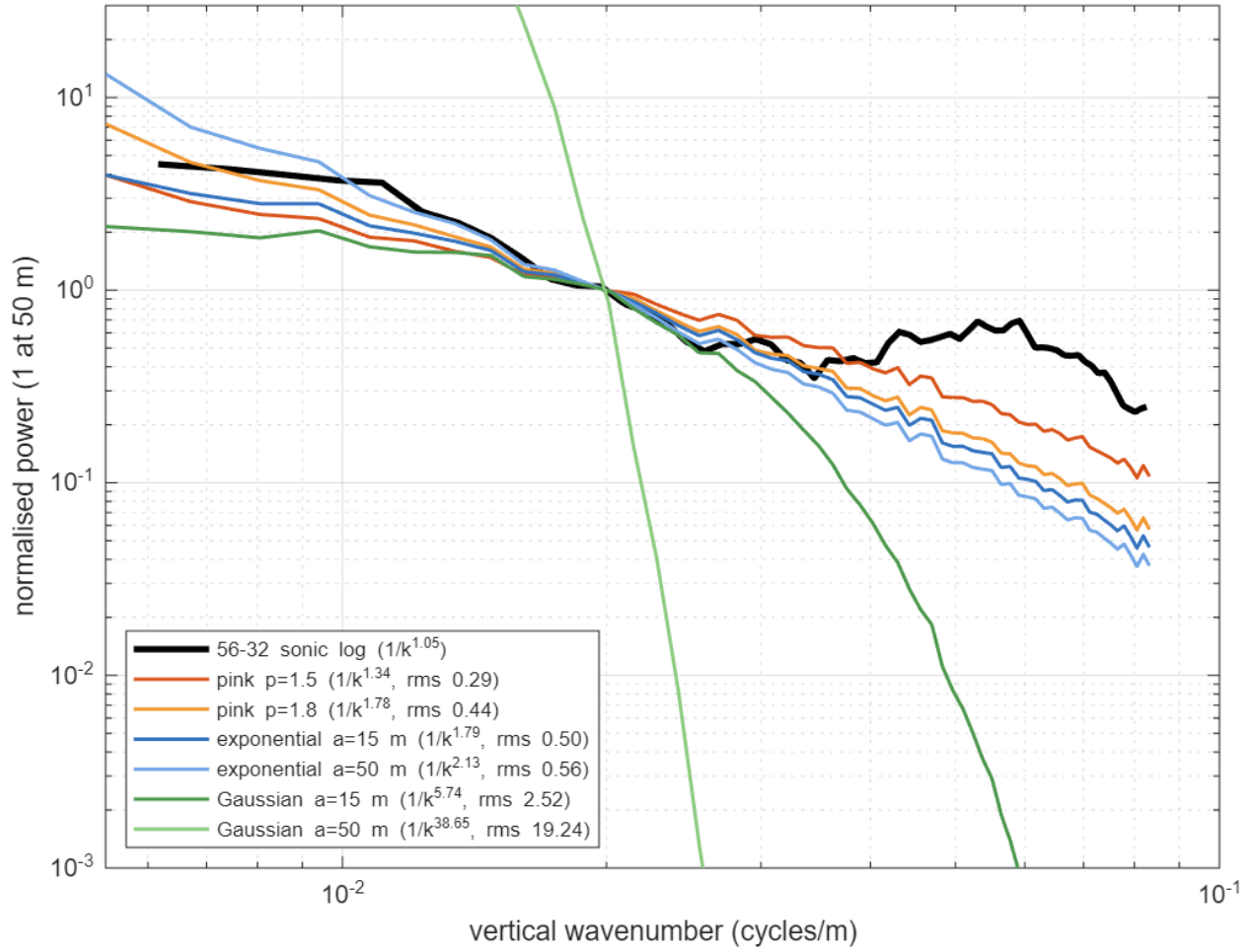


Fig. S11 Vertical fluctuation spectra of the 56-32 sonic log (black) and of the model media over 12–200 m, normalised at 50 m. The rms log difference in the legend is a descriptive distance, not a likelihood.

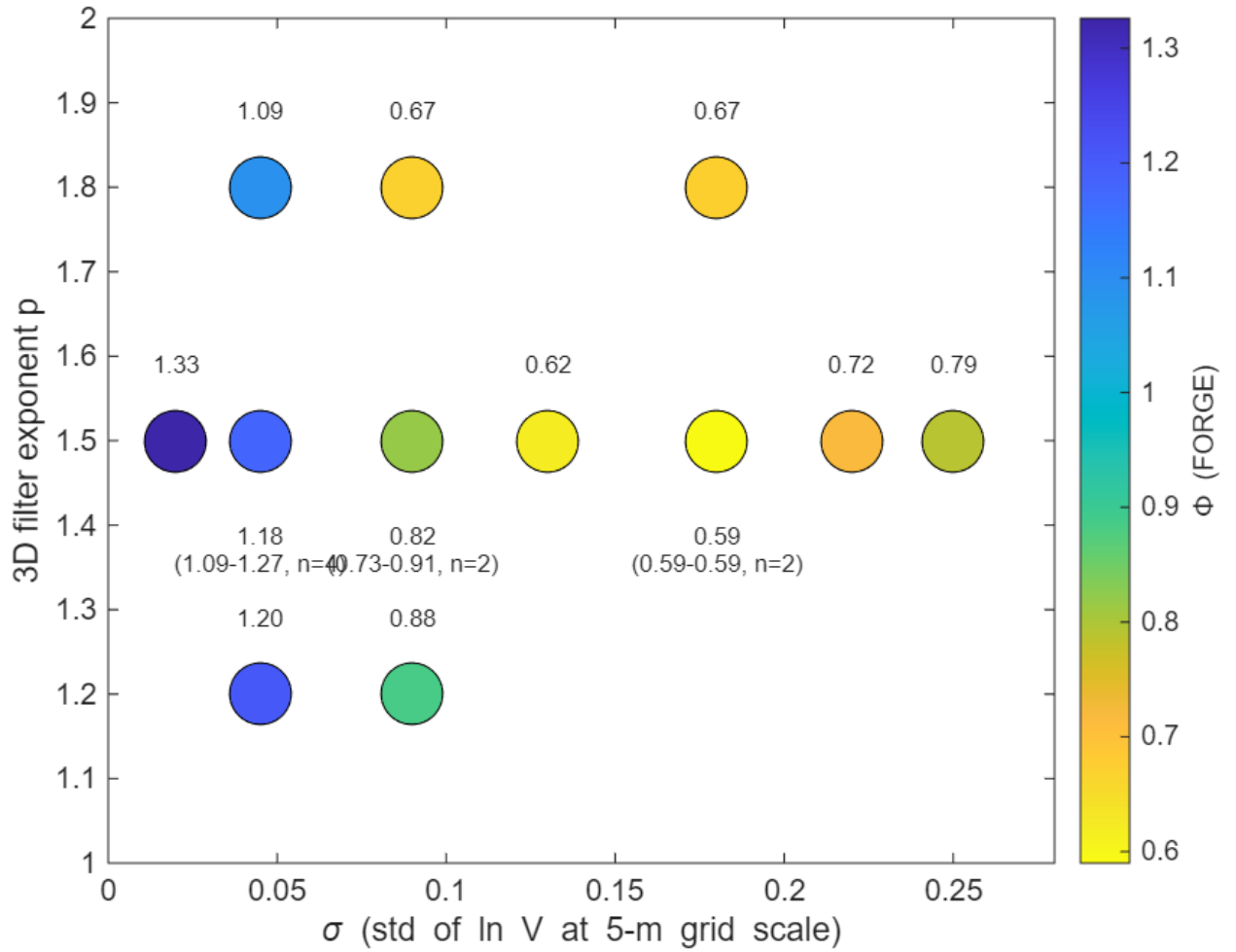


Fig. S12 Original Φ over the pink-family parameters (σ , p) for FORGE; mean and range where several realisations exist. $\sigma = 0.22$ and 0.25 have fewer than five grid points per minimum S wavelength at 80 Hz.

Supplementary tables

Table S1 Window sensitivity of the per-event coda test

Fit end (ms)	Coda (ms)	H	L	L6	GRAD 3	FZ	A	P1	P3	P6	G50
S + 15	S + 40–100	–1.80/– 1.09	–1.64/– 1.17	–1.28/– 0.53	–1.67/– 1.11	–1.04/– 0.39	–1.91/– 1.19	–0.74/+ 0.02	–0.14/+ 0.53	+0.31/+ 0.78	+0.29/+ 0.54
S + 15	S + 50–110	–2.21/– 1.41	–1.99/– 1.39	–1.46/– 0.64	–2.11/– 1.44	–1.33/– 0.65	–2.15/– 1.42	–0.90/– 0.08	–0.30/+ 0.50	+0.14/+ 0.76	+0.16/+ 0.49
S + 15	S + 30–100	–1.58/– 0.99	–1.45/– 1.03	–1.11/– 0.48	–1.49/– 1.02	–0.82/– 0.19	–1.55/– 1.05	–0.60/+ 0.05	–0.03/+ 0.45	+0.39/+ 0.72	+0.35/+ 0.55
S + 10	S + 40–100	–1.77/– 1.07	–1.64/– 1.16	–1.29/– 0.52	–1.65/– 1.09	–1.04/– 0.33	–1.91/– 1.20	–0.59/+ 0.03	–0.07/+ 0.58	+0.33/+ 0.88	+0.31/+ 0.59
S + 10	S + 50–110	–2.17/– 1.41	–1.99/– 1.40	–1.44/– 0.64	–2.09/– 1.43	–1.32/– 0.61	–2.15/– 1.42	–0.78/– 0.08	–0.22/+ 0.52	+0.17/+ 0.84	+0.16/+ 0.54
S + 10	S + 30–100	–1.57/– 0.96	–1.46/– 1.03	–1.12/– 0.48	–1.47/– 0.99	–0.82/– 0.16	–1.58/– 1.05	–0.50/+ 0.04	+0.04/+ 0.52	+0.41/+ 0.84	+0.37/+ 0.58
S + 20	S + 40–100	–1.79/– 1.08	–1.64/– 1.18	–1.28/– 0.53	–1.67/– 1.11	–1.07/– 0.39	–1.91/– 1.19	–0.72/– 0.00	–0.15/+ 0.50	+0.28/+ 0.78	+0.26/+ 0.50
S + 20	S + 50–110	–2.21/– 1.41	–2.00/– 1.41	–1.45/– 0.62	–2.11/– 1.44	–1.32/– 0.67	–2.15/– 1.42	–0.90/– 0.10	–0.34/+ 0.48	+0.14/+ 0.75	+0.11/+ 0.45
S + 20	S + 30–100	–1.58/– 0.98	–1.46/– 1.04	–1.11/– 0.47	–1.49/– 1.02	–0.85/– 0.20	–1.55/– 1.03	–0.58/+ 0.02	–0.02/+ 0.42	+0.36/+ 0.68	+0.31/+ 0.50
S + 60	S + 40–100	–1.79/– 1.09	–1.65/– 1.18	–1.28/– 0.55	–1.67/– 1.13	–1.13/– 0.48	–1.89/– 1.20	–0.86/– 0.17	–0.34/+ 0.28	+0.10/+ 0.50	+0.16/+ 0.25
S + 60	S + 50–110	–2.22/– 1.41	–1.98/– 1.40	–1.45/– 0.63	–2.12/– 1.45	–1.40/– 0.74	–2.14/– 1.42	–1.04/– 0.21	–0.48/+ 0.28	+0.00/+ 0.53	+0.07/+ 0.16
S + 60	S + 30–100	–1.59/–	–1.45/–	–1.12/–	–1.50/–	–0.93/–	–1.54/–	–0.73/–	–0.21/+	+0.23/+	+0.22/+

100	0.99	1.03	0.50	1.03	0.29	1.03	0.15	0.25	0.44	0.22
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Median $\log_{10}(\text{predicted/observed late energy})$, sensor A/B. The fit window starts at $P - 10$ ms. Rows with fit end $S + 60$ ms overlap the coda window (original analysis).

Table S2 Moment-tensor variants: coda prediction, fit, conditioning and decomposition

Medium	Full A/B	Deviatoric A/B	DC A/B	VR full	VR dev	VR DC	Cond. median (p90)	ISO / CLVD /DC (%)
H	-1.80/-1.09	-1.83/-1.09	-1.82/-1.11	0.75	0.71	0.67	17.6 (19.3)	28/28/41
L	-1.64/-1.17	-1.70/-1.16	-1.71/-1.18	0.73	0.68	0.64	17.4 (18.8)	29/31/38
L6	-1.28/-0.53	-1.33/-0.51	-1.34/-0.51	0.73	0.68	0.66	17.8 (19.4)	29/35/37
GRAD3	-1.67/-1.11	-1.69/-1.11	-1.69/-1.13	0.74	0.69	0.66	17.4 (19.0)	28/26/42
FZ	-1.04/-0.39	-1.19/-0.49	-1.24/-0.58	0.73	0.69	0.65	15.4 (18.5)	30/27/41
A	-1.91/-1.19	-2.04/-1.21	-2.10/-1.25	0.64	0.59	0.54	16.4 (24.7)	28/35/36
P1	-0.74/+0.02	-0.91/-0.16	-0.93/-0.29	0.72	0.64	0.61	12.0 (17.0)	34/28/36
P3	-0.14/+0.53	-0.22/+0.41	-0.26/+0.35	0.67	0.60	0.57	8.1 (13.6)	36/24/37
P6	+0.31/+0.78	+0.28/+0.73	+0.23/+0.72	0.62	0.55	0.52	6.6 (11.1)	35/22/41
E15	+0.28/+0.94	+0.34/+0.88	+0.31/+0.84	0.64	0.55	0.51	6.2 (9.5)	34/23/39
E50	+0.31/+0.71	+0.24/+0.69	+0.23/+0.64	0.59	0.54	0.50	6.5 (10.6)	30/21/42
G15	+0.42/+1.12	+0.41/+1.03	+0.39/+0.98	0.64	0.55	0.52	6.5 (9.5)	36/22/37
G50	+0.29/+0.54	+0.30/+0.57	+0.30/+0.52	0.56	0.51	0.48	5.7 (8.1)	31/24/41
VSD	-0.13/+0.54	-0.14/+0.49	-0.19/+0.46	0.66	0.62	0.58	8.8 (14.7)	25/27/41
VSO	+0.32/+0.79	+0.27/+0.74	+0.22/+0.68	0.62	0.55	0.52	6.3 (11.3)	31/23/42

Median $\log_{10}(\text{pred/obs})$ at the primary windows. Condition number of the column-normalised design matrix. Decomposition of the full solutions (median absolute percentages); with two sensors in one well the non-DC parts are poorly resolved and are not interpreted.

Table S3 Per-event coda test for well-timed event subsets

Medium	All (416)	S peak ± 10 ms (250)	S peak ± 5 ms (168)
H	-1.80/-1.09	-1.64/-0.98	-1.58/-0.98
L	-1.64/-1.17	-1.42/-1.14	-1.35/-1.13
L6	-1.28/-0.53	-1.10/-0.53	-1.04/-0.55
GRAD3	-1.67/-1.11	-1.52/-1.04	-1.41/-1.05
FZ	-1.04/-0.39	-0.88/-0.35	-0.87/-0.42
A	-1.91/-1.19	-1.70/-1.17	-1.62/-1.20
P1	-0.74/+0.02	-0.46/+0.08	-0.36/+0.05
P3	-0.14/+0.53	+0.02/+0.53	+0.08/+0.51
P6	+0.31/+0.78	+0.38/+0.81	+0.40/+0.80
E15	+0.28/+0.94	+0.39/+0.98	+0.40/+0.95
E50	+0.31/+0.71	+0.45/+0.72	+0.52/+0.68
G15	+0.42/+1.12	+0.64/+1.12	+0.70/+1.09
G50	+0.29/+0.54	+0.51/+0.53	+0.52/+0.53

Median $\log_{10}(\text{pred/obs})$, sensor A/B. Subsets: events whose observed 40–80 Hz S-envelope peak lies within the stated interval of the predicted S time on both sensors.

Table S4 Sensor self-calibration: event bootstrap and leave-one-stage-out

Parameter	Estimate	Bootstrap s.d.	95 % interval	Without stage 2	Without stage 3
East (m)	833.4	2.3	[827.7, 836.7]	826.0	834.0
North (m)	-5.9	3.1	[-11.0, 2.2]	-9.9	9.0
Depth A (m)	2179.5	6.1	[2166.5, 2191.4]	2182.1	2231.4
Depth B (m)	2522.5	1.9	[2519.4, 2526.3]	2519.4	2573.9
Clock A (ms)	-5.0	0.9	[-7.5, -3.7]	-6.7	-7.6
Clock B (ms)	-3.0	0.6	[-4.5, -2.3]	-4.4	-11.7
V_P (m/s)	5808.3	42.7	[5710.2, 5873.0]	5653.1	5372.6

Event bootstrap (500 resamples). Leaving out stage 3 (most events) shifts both sensors ~ 50 m deeper and lowers V_P by 8 %: depth, clock and velocity trade off with the catalogue locations, so the bootstrap intervals understate the true uncertainty.

Table S5 Apparent corner frequencies of S-window displacement spectra

Sensor	Magnitude	Events	% corner ≤ 300 Hz	Median corner (Hz)	% corner < 80 Hz	Slope 30–150 Hz
A	all	550	99	128	21	-0.45
A	-1 to -0.3	381	99	143	16	-0.21
A	-0.3 to 0.3	165	100	100	31	-0.75
A	≥ 0.3	4	100	87	50	-1.27
B	all	533	85	90	34	-0.52
B	-1 to -0.3	366	82	90	34	-0.39
B	-0.3 to 0.3	163	92	90	35	-0.74
B	≥ 0.3	4	100	58	75	-1.28

Apparent corner: first frequency above 50 Hz at which the smoothed displacement spectrum falls 6 dB below its 30–50 Hz level (searched to 300 Hz, below the resonances). It includes path attenuation and is therefore a lower bound on the source corner. Slope: log–log displacement-spectrum slope where amplitude $\text{SNR} > 3$.

Table S6 Sensitivity of synthetic late energy and S width to source duration

Medium	Brune corner (Hz)	Sensor	$\log_{10}$ coda/S	S width (ms)
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H	impulsive	A	-3.09	8.9
H	impulsive	B	-3.06	9.1
H	150	A	-2.95	9.0
H	150	B	-2.92	9.2
H	100	A	-2.87	9.2
H	100	B	-2.84	9.4
H	60	A	-2.74	9.4
H	60	B	-2.71	9.6
P6	impulsive	A	-0.98	12.3
P6	impulsive	B	-1.50	10.9
P6	150	A	-0.95	12.4
P6	150	B	-1.48	11.0
P6	100	A	-0.94	12.5
P6	100	B	-1.47	11.1
P6	60	A	-0.95	12.7
P6	60	B	-1.44	11.2

Homogeneous (H) and pink $\sigma = 0.13$ (P6) synthetics with double-couple sources, convolved with Brune moment-rate functions; median over events; windows as in the main text.

Table S7 Ringing tests: spectral-peak prominence and null-ringing deficits

Sensor / test	Band / resonance	S-spectrum peak prominence (dB)	Coda/S ratio peak prominence (dB)
A	30-120	0.30	0.17
A	280-380	0.29	0.47
A	420-560	1.50	0.60
A	850-1000	6.47	0.58
B	30-120	0.35	0.36
B	280-380	5.29	1.26
B	420-560	7.86	1.92
B	850-1000	9.60	0.36
null test	f : 40 Hz, Q 5	deficit A -1.99	deficit B -1.33
null test	f : 40 Hz, Q 10	deficit A -1.98	deficit B -1.32
null test	f : 40 Hz, Q 20	deficit A -1.93	deficit B -1.28
null test	f : 40 Hz, Q 40	deficit A -1.91	deficit B -1.26
null test	f : 50 Hz, Q 5	deficit A -1.63	deficit B -0.97
null test	f : 50 Hz, Q 10	deficit A -1.61	deficit B -0.96
null test	f : 50 Hz, Q 20	deficit A -1.67	deficit B -1.02
null test	f : 50 Hz, Q 40	deficit A -1.72	deficit B -1.06
null test	f : 60 Hz, Q 5	deficit A -1.90	deficit B -1.24
null test	f : 60 Hz, Q 10	deficit A -1.88	deficit B -1.23
null test	f : 60 Hz, Q 20	deficit A -1.87	deficit B -1.21
null test	f : 60 Hz, Q 40	deficit A -1.87	deficit B -1.21
null test	f : 70 Hz, Q 5	deficit A -1.85	deficit B -1.18
null test	f : 70 Hz, Q 10	deficit A -1.80	deficit B -1.15
null test	f : 70 Hz, Q 20	deficit A -1.80	deficit B -1.15
null test	f : 70 Hz, Q 40	deficit A -1.82	deficit B -1.16
null test	f : 80 Hz, Q 5	deficit A -1.82	deficit B -1.17
null test	f : 80 Hz, Q 10	deficit A -1.82	deficit B -1.17
null test	f : 80 Hz, Q 20	deficit A -1.84	deficit B -1.18
null test	f : 80 Hz, Q 40	deficit A -1.85	deficit B -1.19

Upper rows: maximum prominence of peaks in the smoothed median S spectrum and coda/S spectral ratio. Lower rows: homogeneous synthetics convolved with a single resonance scaled so that its spectral peak equals the observed 30–120 Hz prominence; deficit = $\log_{10}$ of predicted/observed median coda/S.

Table S8 Observed late energy by stimulation stage

Sensor	Stage	Events	Median $\log_{10}$ coda/S	Median distance (m)
A	2	14	-1.52	484
A	3	402	-1.23	392
B	2	14	-1.62	379
B	3	402	-1.88	300

Stage 2 has only 14 events and differs from stage 3 also in distance; a stage-by-stage test of stimulation-dependent scattering is not possible with these data.

Table S9 Random-field statistics after truncation

Medium	$\sigma(\ln V)$ after truncation	% cells truncated	V_P range (m/s)	V_S range (m/s)
pink s.13 p1.5	0.1297	0.28	3933–8579	2273–4959
pink s.18 p1.5	0.1780	1.25	3704–9109	2141–5265
exp a15 s.13	0.1297	0.28	3933–8579	2273–4959
exp a50 s.13	0.1297	0.25	3933–8579	2273–4959
Gauss a15 s.13	0.1296	0.28	3933–8579	2273–4959
Gauss a50 s.13	0.1297	0.26	3933–8579	2273–4959

Truncation at $\pm 3\sigma$ ($\pm 2.5\sigma$ for $\sigma > 0.15$) without renormalisation; density is constant (2650 kg/m³).

Table S10 Numerical convergence of coda measures

Case	Measure	Band	Median 5-m reference	Median case	Median difference	Robust s.d. of difference	Events
C25	$\log_{10}$ coda/S (S+40-100 ms)	20-40 Hz	-1.4	-1.23	+0.156	0.127	358
C25	$\log_{10}$ coda/S (S+40-100 ms)	40-80 Hz	-1.41	-1.42	+0.007	0.109	358

C25	coda level (log10)	20-40 Hz	-0.704	-0.67	+0.036	0.044	358
C25	coda level (log10)	40-80 Hz	-0.945	-0.953	+0.002	0.076	358
C25	log10 Qc^-1	20-40 Hz	-0.699	-0.749	+0.057	0.144	358
C25	log10 Qc^-1	40-80 Hz	-1.02	-0.991	+0.026	0.072	358
C25	S width (ms)	20-40 Hz	16.5	16.5	+0.022	0.047	358
C25	S width (ms)	40-80 Hz	14.5	14.6	-0.004	0.228	358
C25	decay rate (1/s)	20-40 Hz	49	45.4	-3.082	4.920	358
C25	decay rate (1/s)	40-80 Hz	48.4	50.8	+2.141	5.484	358
CBIG	log10 coda/S (S+40-100 ms)	20-40 Hz	-1.4	-1.27	+0.090	0.279	358
CBIG	log10 coda/S (S+40-100 ms)	40-80 Hz	-1.41	-0.982	+0.403	0.240	358
CBIG	coda level (log10)	20-40 Hz	-0.704	-0.687	+0.011	0.097	358
CBIG	coda level (log10)	40-80 Hz	-0.945	-0.726	+0.187	0.175	358
CBIG	log10 Qc^-1	20-40 Hz	-0.699	-0.983	-0.262	0.239	358
CBIG	log10 Qc^-1	40-80 Hz	-1.02	-1.68	-0.669	0.522	358
CBIG	S width (ms)	20-40 Hz	16.5	16.4	-0.023	0.065	358
CBIG	S width (ms)	40-80 Hz	14.5	14.6	+0.023	0.194	358
CBIG	decay rate (1/s)	20-40 Hz	49	32.4	-14.958	11.486	358
CBIG	decay rate (1/s)	40-80 Hz	48.4	20.6	-26.329	13.229	358
CSP	log10 coda/S (S+40-100 ms)	20-40 Hz	-1.4	-1.5	-0.131	0.210	358
CSP	log10 coda/S (S+40-100 ms)	40-80 Hz	-1.41	-1.27	+0.125	0.135	358
CSP	coda level (log10)	20-40 Hz	-0.704	-0.74	-0.035	0.061	358
CSP	coda level (log10)	40-80 Hz	-0.945	-0.904	+0.052	0.082	358
CSP	log10 Qc^-1	20-40 Hz	-0.699	-0.727	+0.076	0.158	358
CSP	log10 Qc^-1	40-80 Hz	-1.02	-1.12	-0.095	0.084	358
CSP	S width (ms)	20-40 Hz	16.5	16.4	-0.034	0.050	358
CSP	S width (ms)	40-80 Hz	14.5	14.6	+0.001	0.125	358
CSP	decay rate (1/s)	20-40 Hz	49	46.8	-1.337	8.200	358
CSP	decay rate (1/s)	40-80 Hz	48.4	41	-6.333	4.949	358

C25: 2.5-m grid; CBIG: domain enlarged by 150 m on every side; CSP: sponge of 40 instead of 20 nodes. Same band-limited medium ($\sigma = 0.13$), sensor B, events and double-couple mechanisms as the 5-m reference C5. Differences are per event.

Table S11 Median predicted/observed late energy versus σ , and σ_{eff}

Family	σ	Realisations	Median $\log_{10}(\text{pred/obs})$, A	Median, B
pink15	0.020	1	-1.20	-0.49
pink15	0.045	4	-0.75	-0.11
pink15	0.090	2	-0.19	+0.38
pink15	0.130	2	+0.29	+0.59
pink15	0.180	2	+0.71	+0.85
pink18	0.045	1	-0.73	-0.03
pink18	0.090	2	-0.12	+0.28
pink18	0.130	2	+0.38	+0.53
pink18	0.180	2	+0.90	+0.85
exp15	0.090	2	-0.04	+0.50
exp15	0.130	2	+0.34	+0.79
exp15	0.180	2	+0.77	+1.06
exp50	0.090	2	-0.20	+0.16
exp50	0.130	2	+0.38	+0.44
exp50	0.180	2	+0.98	+0.91
gau15	0.090	2	+0.18	+0.70
gau15	0.130	2	+0.49	+1.00
gau15	0.180	2	+0.82	+1.28
gau50	0.090	2	-0.47	-0.58
gau50	0.130	2	+0.33	-0.02
gau50	0.180	2	+0.95	+0.70
pink15	σ_{eff}		0.104	0.053
pink18	σ_{eff}		0.098	0.048
exp15	σ_{eff}		0.093	outside grid
exp50	σ_{eff}		0.102	outside grid
gau15	σ_{eff}		outside grid	outside grid
gau50	σ_{eff}		0.112	0.131

Averaged over realisations; σ_{eff} interpolated linearly in $\ln \sigma$ at the zero crossing.

Table S12 Statistical misfit Φ with event-bootstrap 95 % intervals

Medium	Family	σ	p or a (m)	Φ	95 % interval
X exp 0.13 50 33	exp50	0.130	50	0.246	[0.237, 0.297]
X pink 0.13 1.8 33	pink	0.130	1.8	0.291	[0.273, 0.339]
X exp 0.09 50 11	exp50	0.090	50	0.371	[0.347, 0.432]
P8	pink	0.090	1.8	0.397	[0.377, 0.451]
R6	pink	0.130	1.5	0.400	[0.370, 0.459]
G50	gau50	0.130	50	0.402	[0.377, 0.459]
E15b	exp15	0.130	15	0.418	[0.394, 0.467]
X pink 0.13 1.8 11	pink	0.130	1.8	0.455	[0.429, 0.512]
E50	exp50	0.130	50	0.459	[0.431, 0.516]
VSD	pink-VpVs	0.130	1.5	0.468	[0.437, 0.529]
P3	pink	0.090	1.5	0.469	[0.440, 0.522]
VSO	pink-VpVs	0.130	1.5	0.476	[0.457, 0.531]
P6	pink	0.130	1.5	0.483	[0.460, 0.536]
X exp 0.09 15 11	exp15	0.090	15	0.496	[0.466, 0.541]

R5	pink	0.180	1.5	0.522	[0.480, 0.581]
G50b	gau50	0.130	50	0.523	[0.493, 0.591]
X gau 0.09 50 11	gau50	0.090	50	0.535	[0.491, 0.610]
E15	exp15	0.130	15	0.551	[0.522, 0.598]
X gau 0.09 15 11	gau15	0.090	15	0.569	[0.539, 0.617]
X gau 0.13 15 33	gau15	0.130	15	0.572	[0.541, 0.619]
X exp 0.09 15 33	exp15	0.090	15	0.577	[0.535, 0.645]
X gau 0.09 15 33	gau15	0.090	15	0.598	[0.559, 0.665]
X exp 0.09 50 33	exp50	0.090	50	0.608	[0.552, 0.680]
X pink 0.09 1.8 33	pink	0.090	1.8	0.609	[0.559, 0.678]
R3	pink	0.090	1.5	0.613	[0.570, 0.673]
X exp 0.18 15 33	exp15	0.180	15	0.627	[0.579, 0.690]
P9	pink	0.180	1.5	0.640	[0.605, 0.697]
X gau 0.18 50 33	gau50	0.180	50	0.646	[0.590, 0.726]
X pink 0.18 1.8 33	pink	0.180	1.8	0.647	[0.597, 0.712]
G15	gau15	0.130	15	0.652	[0.621, 0.705]
X exp 0.18 50 33	exp50	0.180	50	0.679	[0.626, 0.746]
P11	pink	0.180	1.8	0.732	[0.684, 0.797]
X exp 0.18 15 11	exp15	0.180	15	0.733	[0.691, 0.802]
X gau 0.18 50 11	gau50	0.180	50	0.733	[0.685, 0.804]
X gau 0.18 15 33	gau15	0.180	15	0.745	[0.706, 0.802]
X gau 0.18 15 11	gau15	0.180	15	0.757	[0.735, 0.817]
X exp 0.18 50 11	exp50	0.180	50	0.761	[0.719, 0.825]
X gau 0.09 50 33	gau50	0.090	50	0.863	[0.814, 0.939]
L6	deterministic			1.014	[0.962, 1.088]
FZ	deterministic			1.165	[1.113, 1.227]
L	smooth			1.328	[1.263, 1.403]
A	smooth			1.356	[1.293, 1.425]
GRAD3	smooth			1.459	[1.403, 1.526]
H	smooth			1.466	[1.406, 1.532]

200 event-bootstrap resamples (identical for every medium). Φ as defined in the main text.

Table S13 Family comparison: best member, bootstrap and holdout

Family	Best Φ	Bootstrap median [95 %]	% bootstrap wins	Held-out Φ median [range]
exp50	0.246	0.264 [0.237, 0.297]	98	0.265 [0.242, 0.353]
pink	0.291	0.306 [0.273, 0.339]	2	0.307 [0.273, 0.399]
gau50	0.402	0.414 [0.377, 0.459]	0	0.416 [0.381, 0.709]
exp15	0.418	0.429 [0.394, 0.467]	0	0.427 [0.383, 0.526]
pink-VpVs	0.468	0.472 [0.437, 0.512]	0	0.498 [0.453, 0.670]
gau15	0.569	0.571 [0.538, 0.606]	0	0.589 [0.537, 0.669]
deterministic	1.014	1.020 [0.962, 1.088]	0	1.022 [0.941, 1.143]
smooth	1.328	1.333 [1.263, 1.403]	0	1.336 [1.247, 1.429]

Best member per bootstrap resample; holdout: best (σ , shape, seed) chosen on a training half and scored on the other half (20 random splits + near/far split, both directions).

Table S14 Equal- σ family test ($\sigma = 0.13$): family scores for nine synthetic truths

Truth	Pink	Exponential ($a = 15$ m)	Gaussian ($a = 50$ m)	Selected
pink (R6)	0.25	0.34	0.46	pink ✓
pink (R6c)	0.27	0.21	0.35	exponential ✗
pink (R6d)	0.30	0.40	0.53	pink ✓
exponential (E15b)	0.21	0.24	0.34	pink ✗
exponential (E15c)	0.35	0.24	0.34	exponential ✓
exponential (E15d)	0.21	0.26	0.41	pink ✗
Gaussian (G50b)	0.39	0.43	0.36	Gaussian ✓
Gaussian (G50c)	0.45	0.36	0.30	Gaussian ✓
Gaussian (G50d)	0.41	0.43	0.53	pink ✗
FORGE data (original set)	0.61	0.54	0.51	—

Family score = mean Φ over candidate realisations whose seed differs from the truth. 5 of 9 correct; $P(\geq 5 \text{ of } 9 \mid p = 1/3) = 0.145$. Pink and exponential fields built from the same seed are strongly correlated.

Table S15 Sonic-log spectral slope and spectral-model comparison

Detrend / band	PSD / model	Segment / parameter	β (12–200 m) / rms	β (1–200 m) / ΔAIC
mean	periodogram	full	1.04	1.54
mean	welch	full	0.87	1.51
mean	multitaper	full	0.84	1.55
mean	multitaper	third 1 (1950–2221 m)	0.85	1.20
mean	multitaper	third 2 (2221–2493 m)	0.91	1.70
mean	multitaper	third 3 (2493–2764 m)	1.21	1.68
linear	periodogram	full	1.05	1.54
linear	welch	full	0.87	1.51
linear	multitaper	full	0.85	1.55

linear	multitaper	third 1 (1950-2221 m)	0.59	1.18
linear	multitaper	third 2 (2221-2493 m)	0.87	1.70
linear	multitaper	third 3 (2493-2764 m)	0.95	1.67
quadratic	periodogram	full	1.05	1.54
quadratic	welch	full	0.87	1.51
quadratic	multitaper	full	0.84	1.55
quadratic	multitaper	third 1 (1950-2221 m)	0.57	1.18
quadratic	multitaper	third 2 (2221-2493 m)	0.83	1.70
quadratic	multitaper	third 3 (2493-2764 m)	0.78	1.66
12-200 m	power law		rms 0.302	Δ AIC 0.0
12-200 m	exponential ACF (Lorentzian)	a = 2.3 m	rms 0.344	Δ AIC 3.4
12-200 m	Gaussian ACF	a = 3.6 m	rms 0.348	Δ AIC 3.7
12-200 m	power law with corner	a = 177032.3 m	rms 0.302	Δ AIC 2.0
1-200 m	power law		rms 0.454	Δ AIC 0.0
1-200 m	exponential ACF (Lorentzian)	a = 2.3 m	rms 0.483	Δ AIC 2.2
1-200 m	Gaussian ACF	a = 0.8 m	rms 0.928	Δ AIC 25.7
1-200 m	power law with corner	a = 8.2 m	rms 0.447	Δ AIC 1.4

Upper rows: least-squares slope of $\ln$ PSD versus $\ln k$ (unbinned). Lower rows: fits to log-binned multitaper spectra (equal weight per log bin), which give a flatter power-law slope over 12–200 m (0.54) than the unbinned fit; the slope over this 1.2-decade band is poorly determined.

Table S16 Per-event coda test for every medium and seed

Medium	Median A/B (all 416)	% within $\times 3$ A/B	VR	Median A/B (150 highest SNR)
A	-1.91/-1.19	1/11	0.64	-1.86/-1.26
E15	+0.28/+0.94	50/22	0.64	+0.36/+0.84
E15b	+0.40/+0.64	37/29	0.68	+0.57/+0.58
E50	+0.31/+0.71	45/27	0.59	+0.32/+0.55
FZ	-1.04/-0.39	24/45	0.73	-0.77/-0.33
G15	+0.42/+1.12	43/15	0.64	+0.61/+1.05
G50	+0.29/+0.54	45/33	0.56	+0.26/+0.47
G50b	+0.36/-0.58	34/38	0.65	+0.68/-0.69
GRAD3	-1.67/-1.11	3/9	0.74	-1.61/-1.12
H	-1.80/-1.09	2/9	0.75	-1.74/-1.09
L	-1.64/-1.17	5/5	0.73	-1.53/-1.21
L6	-1.28/-0.53	15/41	0.73	-1.22/-0.65
P1	-0.74/+0.02	34/51	0.72	-0.55/-0.05
P11	+0.77/+1.06	29/15	0.56	+0.72/+0.96
P2	-0.73/-0.03	31/54	0.71	-0.61/-0.13
P3	-0.14/+0.53	50/35	0.67	+0.00/+0.43
P4	-1.20/-0.49	16/42	0.73	-1.08/-0.52
P5	-0.74/-0.01	32/49	0.72	-0.60/-0.12
P6	+0.31/+0.78	50/26	0.62	+0.34/+0.70
P8	-0.06/+0.50	51/36	0.66	+0.02/+0.41
P9	+0.61/+1.03	37/18	0.60	+0.57/+0.92
R1	-0.65/-0.08	32/51	0.73	-0.45/-0.17
R2	-0.72/-0.04	36/50	0.73	-0.51/-0.16
R3	-0.23/+0.24	43/45	0.72	-0.04/+0.05
R5	+0.82/+0.68	31/28	0.64	+0.94/+0.63
R6	+0.26/+0.40	47/40	0.69	+0.39/+0.22
T1	-0.89/-0.34	27/46	0.77	-0.68/-0.38
VSD	-0.13/+0.54	53/35	0.66	-0.13/+0.38
VSO	+0.32/+0.79	49/24	0.62	+0.41/+0.68
X exp 0.09 15 11	-0.07/+0.61	51/33	0.68	+0.01/+0.48
X exp 0.09 15 33	-0.00/+0.40	42/41	0.71	+0.24/+0.27
X exp 0.09 50 11	-0.16/+0.39	48/41	0.65	-0.12/+0.30
X exp 0.09 50 33	-0.25/-0.06	40/50	0.72	-0.07/-0.15
X exp 0.13 50 33	+0.45/+0.18	34/47	0.67	+0.63/+0.11
X exp 0.18 15 11	+0.70/+1.26	34/13	0.60	+0.71/+1.11
X exp 0.18 15 33	+0.84/+0.86	30/24	0.64	+1.10/+0.83
X exp 0.18 50 11	+0.81/+1.24	29/13	0.55	+0.80/+1.12
X exp 0.18 50 33	+1.16/+0.57	22/33	0.60	+1.34/+0.46
X gau 0.09 15 11	+0.14/+0.79	48/22	0.69	+0.28/+0.76
X gau 0.09 15 33	+0.23/+0.62	45/36	0.71	+0.41/+0.52
X gau 0.09 50 11	-0.45/-0.35	39/44	0.62	-0.45/-0.54
X gau 0.09 50 33	-0.50/-0.80	26/30	0.71	-0.28/-1.00
X gau 0.13 15 33	+0.56/+0.89	36/22	0.67	+0.66/+0.75
X gau 0.18 15 11	+0.72/+1.37	32/10	0.60	+0.79/+1.28
X gau 0.18 15 33	+0.91/+1.19	26/11	0.61	+1.01/+1.10
X gau 0.18 50 11	+0.85/+1.48	25/10	0.54	+0.86/+1.44
X gau 0.18 50 33	+1.05/-0.08	25/53	0.56	+1.33/-0.21
X pink 0.09 1.8 33	-0.19/+0.06	41/51	0.71	-0.03/-0.09
X pink 0.13 1.8 11	+0.34/+0.76	43/28	0.61	+0.36/+0.64
X pink 0.13 1.8 33	+0.43/+0.29	36/45	0.65	+0.50/+0.19
X pink 0.18 1.8 33	+1.04/+0.64	24/32	0.59	+1.30/+0.60

Primary windows (fit P = 10 ms to S + 15 ms; coda S + 40 to S + 100 ms). Common-grid media are named X_<family>_< σ >_<p or a>_<seed>.